

EECE 571Q

Lecture 03: *Just stay still!*

Tuesday 19 May 2026

Announcements

- Quiz 1 today
- Projects: aiming for 4 groups of 2

Last time

We discussed a variety of possible errors and hardware metrics.

We defined quantum error-correcting codes.

Source: Gottesman, Def. 2.1

"A quantum error-correcting code $(U, \mathcal{E})$ (or QECC for short) is a **partial isometry** $U : \mathcal{H}_K \rightarrow \mathcal{H}_N$ with the set of **correctable errors** $\mathcal{E}$ (consisting of linear maps $E : \mathcal{H}_N \rightarrow \mathcal{H}_M$) with the following property: $\exists$ quantum operation $\mathcal{D} : \mathcal{H}_M \rightarrow \mathcal{H}_K$ such that $\forall E \in \mathcal{E}, \forall |\psi\rangle \in \mathcal{H}_K,$ "

$$\mathcal{D}(EU|\psi\rangle \langle\psi| U^\dagger E^\dagger) = c(E, |\psi\rangle) |\psi\rangle \langle\psi|$$

Last time

We developed a QECC and associated circuitry to correct a single bit flip error, and determined conditions when this was worthwhile.

Learning outcomes

- ➊ Design circuits to detect and correct bit flip, phase flip, and arbitrary errors
- ➋ Outline the conditions under which errors can be corrected
- ➌ Define the Pauli group and state its important properties
- ➍ Express quantum error correcting codes in the stabilizer formalism
- ➎ Characterize whether Pauli errors are correctable by a code using its stabilizer generators

Phase flip code: encoding circuit

Main idea: make phase flip errors “look like” bit flip errors.

Conditions for quantum error correction

How do we know if a particular error is correctable by a QECC?

Sufficient conditions: non-degenerate code; errors map to orthogonal subspaces.

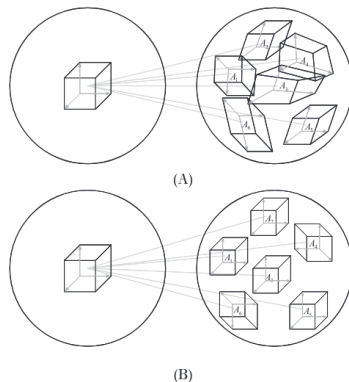


Image: Nielsen and Chuang

Conditions for quantum error correction

Formal statement:

“($Q, \mathcal{E}$) is a QECC iff $\forall |\psi\rangle, |\phi\rangle \in Q, \forall E_a, E_b \in \mathcal{E}$,

Note that C_{ab} does not depend on $|\psi\rangle$ or $|\phi\rangle$ ”

Source: Gottesman, Thm. 2.7

Correcting arbitrary errors

Fact: QECCs are linear.

$$\mathcal{D}(EU|\psi\rangle\langle\psi|U^\dagger E^\dagger) = c(E, |\psi\rangle)\langle\psi|$$

If E_1 and E_2 are correctable, then

$E_1 E_2$ is correctable. (Supplementary exercise; see Gottesman 2.4)

Consequence: if we can correct Pauli errors $\mathcal{E} = \{I, X, Z, XZ\}$, we can correct *arbitrary* errors.

Correcting arbitrary errors

Bit flip, phase flip: obviously Paulis

Coherent errors?, e.g., slight over-rotation by angle θ .

Amplitude damping? *See supplementary exercises*

Shor code

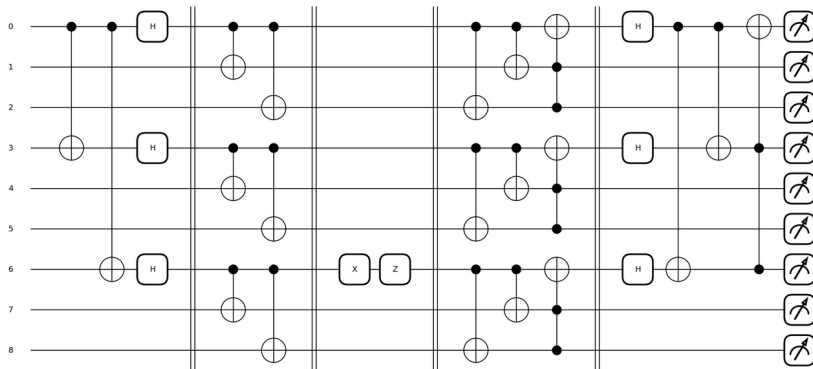
To correct a combination of one bit flip and/or phase flip error, we can *concatenate* codes.

Shor code

The Shor code can correct one *arbitrary error on a single qubit*.

Imagine a bit + phase flip on qubit 6:

Shor code



What about errors on *multiple* qubits?

Source: Gottesman 1.3.2-1.3.4 (see 1.3.4 for proofs)

One option: model as *independent channels*, i.e.,

Revisit our bit flip channel...

If we had $\Phi_{BF} \otimes \Phi_{BF} \otimes \Phi_{BF}$ and p is small, many terms contribute very little.

Often more convenient to model using t -qubit error channels.

What about errors on *multiple* qubits?

Source: Gottesman 1.3.3 - see section for proofs

A linear operator has **weight** t if it acts non-trivially on t qubits tensored with identity on all other qubits.

A linear operator is a **t -qubit error** if it has the form

and each E_i has weight at most t (not necessarily on same qubits).

An n -qubit channel is a **t -qubit error channel** if it has a Kraus decomposition where all Kraus operators are t -qubit errors.

Code distance

The *distance* of a code is the minimum weight, d , of an error F such that the QECC conditions are violated, i.e.,

(for all possible pairs of codewords).

If E_a , E_b are weight t errors, F has weight at most $2t$.

Consequence: a distance d code

- corrects up to $\lfloor (d-1)/2 \rfloor$ errors
- detects up to $d-1$ errors

Code distance

The Shor code is an $((n, k, d)) = ((9, 1, 3))$ code.

Example: find a minimum-weight F error for 9-qubit code that doesn't satisfy criteria.

The Pauli group

The n -qubit Pauli group is

We often ignore the phases:

We'll use shorthand: $X \otimes Z \otimes I \otimes Y = X_0 Z_1 Y_3$.

Exercise: how many elements are in $\mathcal{P}_n$ and $\hat{\mathcal{P}}_n$?

Pauli group

The Pauli group is my favourite group. It has many cool properties.

Fact: any $P_1, P_2 \in \mathcal{P}_n$ either commute or anticommute

Fact: the eigenvalues of all Paulis are ± 1 . Moreover, the eigenspaces of n -qubit Paulis (except I) each have dimension 2^{n-1} .

Prove in supplementary exercises.

Stabilizers

Let T be a QECC. The **stabilizer** of the code, $S(T)$, is

In words: a (sub)group of Paulis for which all codewords are $+1$ eigenstates.

Exercise: Which two-qubit Paulis stabilize this Bell state?

Stabilizers

Facts about $S(T)$:

- $-I$ is never in $S(T)$
- all elements of $S(T)$ commute (Abelian subgroup of $\mathcal{P}_n$)
- choosing a subgroup S defines the stabilized codewords of T

Stabilizer codes

More convenient to work “backwards”: given S , define codespace

If $T = \mathcal{T}(S(T))$, T is a **stabilizer code**.

Bit flip code revisited

An alternative way of thinking about the parity computation is in terms of eigenvalues w.r.t. the code's stabilizer group.

Consider our logical states:

Which three-qubit Paulis are these states $+1$ eigenstates of?

Bit flip code revisited

Let's determine a *minimal* representation of the group in terms of its **stabilizer generators**.

Exercise: if there are r stabilizer generators, how large is the stabilizer group S ?

Stabilizers generators

We can also work backwards to obtain codewords from the stabilizer generators by defining a projector onto the codespace,

Π_S provides insight into size of the stabilized space: each generator cuts space in half, so r generators gives space with dimension 2^{n-r} .

Special case: if $r = n$, have a single state called a *stabilizer state*.

Bit flip code: stabilizers and logical operations

Logical Z : need a Pauli that

- stabilizes the logical state but is *not* in S
- *commutes* with everything in S
- has the right action on the subspace

Logical X : need a Pauli that

- stabilizes the logical state but is *not* in S
- *commutes* with everything in S
- has the right action on the subspace
- anticommutes with logical Z

Phase flip code: stabilizer formalism

Stabilizer measurement and error detection

We can use this formalism to construct circuits for error detection and recovery: simply *measure the stabilizer generators*.

Correctible errors anticommute with at least one generator. We can detect their presence in the syndrome measurement.

Bit flip code: stabilizers and logical errors

Stabilizers and logical errors

Danger: error that *commutes* with all elements of S , but isn't in S .

More precisely, define the *normalizer* $N(S)$ as

For an arbitrary error E ...

- if $E \notin N(S)$, E is detectable
- if $E \in N(S)$ and $E \in S$, the code space is unchanged
- if $E \in N(S)$ but $E \notin S$, E is **undetectable**

Stabilizers and logical errors

Gottesman Thm 3.4

“The set of undetectable errors for a stabilizer code S is $\hat{N}(S) \setminus \hat{S}$. The **distance** of S is $\min\{\text{wt}(E) \mid \hat{E} \in \hat{N}(S) \setminus \hat{S}\}$ ”.

Gottesman Thm 3.5

“The stabilizer code S corrects a set of errors $\mathcal{E} \subseteq \mathcal{P}_n$ iff $\hat{E}^\dagger \hat{F} \notin \hat{N}(S) \setminus \hat{S}$ for all $E, F \in \mathcal{E}$ ”.

Shor code: stabilizer formalism

Stabilizer code usually described by notation $[[n, k, d]]$. The 9-qubit Shor code is a $[[9, 1, 3]]$ code.

Name	Operator
g_1	$ZZIIIIII$
g_2	$IZZIIIIII$
g_3	$III ZZIIII$
g_4	$IIII ZZIIII$
g_5	$IIIIII ZZI$
g_6	$IIIIII IZZ$
g_7	$XXXXXX III$
g_8	$III XXXXXX$
$\tilde{Z}$	$XXXXXXXXXX$
$\tilde{X}$	$ZZZZZZZZZZ$

What is an example of a weight-3 undetectable error?

Screenshot: Nielsen & Chuang, chapter 10.5.6.

Syndromes and decoding

Exercise: What are the error syndromes for Y_1 and $X_0Z_1X_2$?

Name	Operator
g_1	$ZZIIIIIIII$
g_2	$IZZIIIIIIII$
g_3	$III ZZIIIIII$
g_4	$IIII ZZIIII$
g_5	$IIIIII ZZI$
g_6	$IIIIII I ZZ$
g_7	$XXXXXX III$
g_8	$II I XXXXXX$
$\bar{Z}$	$XXXXXXXXXX$
$\bar{X}$	$ZZZZZZZZZZ$

Screenshot: Nielsen & Chuang, chapter 10.5.6.

Syndromes and decoding

Two errors, E and F are in the same **coset** of $N(S)$ if $F = EN$ for some $N \in N(S)$. Errors in the same coset of $N(S)$ have the same error syndrome.

We will explore this more when we discuss decoding.

Next time

Next class:

- Classical linear codes
- CSS codes
- The Steane code

Action items:

- ① Start thinking about project topics
- ② Supplementary exercises

Recommended reading:

- From this class: Gottesman Ch. 3; Codebook EC; N&C 10.5
- For next class: Gottesman Ch. 4; N&C 10.4